

Research Highlights

Manuscript Title: A High-Resolution Finite-Difference Benchmark for Steady Continuation and Hopf Bifurcation Dynamics up to Extreme Reynolds Numbers ($Re = 100,000$) in 2D Lid-Driven Cavity Flows

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- Exact $\mathcal{O}(N^2 \log N)$ DST-I Poisson solver eliminates iterative residual drift.
- Pure central differencing achieves zero numerical viscosity up to $Re = 50,000$.
- Localized hybrid scheme regularizes corner shocks at $Re = 100,000$ on 1025×1025 .
- Asymptotic core vorticity plateau confirms Batchelor's theorem at high Re .
- Time-accurate 513×513 marching resolves limit cycles and unsteady dynamics.